

The Relational Witnessing Framework: The Emergence of Time and Experience from Entanglement Entropy

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Abstract

This paper outlines a formal mathematical framework locating the emergence of time and subjective experience within a globally static, timeless quantum state. Synthesizing the Wheeler–DeWitt equation, Landauer’s Principle, the Page–Wootters mechanism, and the Margolus–Levitin quantum speed limit, we propose that witnessing, entropy, and time are co-generated by a single structural choice—the epistemic bipartition of a timeless whole—through its two canonical maps: the partial trace and conditioning on clock states. The three concepts constitute three projections of the same underlying epistemic cut. Crucially, this framework does not propose new physics; rather, it defines the structural scaffold that any future theory of experience must map from. We formalize the epistemic nature of the partial trace, define the relational step and its entropy increment via a clock–memory decomposition of the witness, derive a relational time bound under an explicit thermodynamic optimality assumption, construct a flag complex encoding the entanglement topology of a witnessing subsystem, and define the categorical and topological constraints that any solution to the hard problem of consciousness must mathematically satisfy.

Methodological note. Axioms I–IV constitute the deductive core of the framework, relying on established physics and information theory; all modeling assumptions on which their consequences depend are stated explicitly and labeled as such. Axiom V and the Topological Hypothesis (Section 6) are conjectures about the form that any bridge between physics and phenomenology must take. We are explicit about where derivation ends and conjecture begins.

Conventions. Throughout this paper, $\log$ denotes the base-2 logarithm and all entropies are measured in *bits*: the von Neumann entropy is $S(\rho) = -\text{Tr}(\rho \log \rho)$, consistent with the base-2 convention used for logarithmic negativity in Section 6. With this convention, Landauer’s principle carries an explicit conversion factor $\ln 2$ (joules per kelvin per bit via $k_B \ln 2$). We further distinguish three conceptually distinct quantities that this convention lets us relate but not conflate: the *entanglement entropy* of a reduced state (an information-theoretic property of a quantum state), the *information-theoretic entropy* of a record (bits erased or overwritten in a memory), and *thermodynamic entropy* (dissipated heat divided by temperature). Where the framework identifies increments of one with increments of another, the identification is stated as an explicit assumption.

1 Axiom I: The Timeless Whole

We begin outside of time. The universe, in its entirety, is a closed system in a pure, static quantum state. There is no external clock, and therefore no objective temporal evolution.

Let the total Hilbert space of the universe be $\mathcal{H}_{\text{total}}$. The universe is described by a single, pure state vector $|\Psi\rangle$. The Hamiltonian $\hat{H}$ of the entire universe yields zero energy, meaning the state is strictly invariant under global time translation [1]:

Axiom 1 (Global Stasis).

$$\hat{H} |\Psi\rangle = 0 \quad (1)$$

Because $|\Psi\rangle$ is a globally pure state, its density matrix $\rho_{\text{total}} = |\Psi\rangle\langle\Psi|$ has von Neumann entropy of exactly zero:

$$S(\rho_{\text{total}}) = -\text{Tr}(\rho_{\text{total}} \log \rho_{\text{total}}) = 0. \quad (2)$$

There is no arrow of time at the global level.

2 Axiom II: The Hilbert Space Bipartition

To yield the emergence of relational dynamics, the timeless whole must be partitioned. We mathematically divide the whole into a Witness (W) and the Rest of the Universe (R) by factoring the Hilbert space via a tensor product decomposition:

Axiom 2 (Bipartition).

$$\mathcal{H}_{\text{total}} = \mathcal{H}_W \otimes \mathcal{H}_R \quad (3)$$

As noted by Zanardi, Lidar, and Lloyd [2], the tensor product structure of a Hilbert space is not uniquely given a priori; it is observable-induced. We adopt the following criterion for the preferred factorization:

Definition 1 (Preferred Factorization). The dynamically preferred bipartition $\mathcal{H}_W \otimes \mathcal{H}_R$ is the tensor product structure with respect to which a specified algebra of observables $\mathcal{A}_W$ acts locally on $\mathcal{H}_W$.

This is a static, algebraic criterion: it selects the factorization by reference to the structure of the observable algebra, not by reference to any dynamical process. For the clock degree of freedom introduced in Axiom IV, the criterion can be strengthened from *preferred* to *unique*: Marletto and Vedral [5] prove that the Page–Wootters clock factorization is free of ambiguity when the clock is non-interacting, i.e., when the global Hamiltonian decomposes as $\hat{H} = \hat{H}_C \otimes \mathbb{I} + \mathbb{I} \otimes \hat{H}_{\text{rest}}$ with no interaction term. We adopt this non-interaction condition for the clock factor in Axiom IV, where its consequences—and an apparent tension with Landauer erasure—are addressed directly (Remark 4).

We note that Zurek’s Quantum Darwinism [3] provides a dynamical confirmation of the algebraic selection in the emergent-time regime—the environment selectively proliferates information about pointer states, stabilizing precisely the factorization that makes $\mathcal{A}_W$ local. However, Quantum Darwinism is not invoked foundationally here, since it presupposes time, which is not defined until Axiom IV. Instead, it serves as a consistency check: the factorization selected algebraically must coincide with the one stabilized dynamically once relational time has emerged.

3 Axiom III: The Epistemic Bipartition

Axiom 3 (Formal Witness). *A subsystem W acts as a formal Witness if and only if it possesses genuine quantum entanglement with R , characterized by:*

- (i) a non-trivial Schmidt rank in the decomposition of $|\Psi\rangle$ across the bipartition, and
- (ii) non-zero logarithmic negativity $E_N(W:R) > 0$.

The requirement of non-zero logarithmic negativity distinguishes genuine quantum entanglement from classical thermodynamic mixedness. A subsystem whose reduced state is mixed solely due to classical ignorance (a separable state) does not qualify as a formal Witness.

Remark 1. For the bipartition of a pure global state $|\Psi\rangle$, conditions (i) and (ii) are equivalent: non-trivial Schmidt rank implies non-zero logarithmic negativity, and vice versa. This follows because Schmidt rank greater than one for a pure state implies the reduced state ρ_W is mixed, and for reductions of pure bipartite states, non-trivial mixedness implies positive partial-transpose negativity. Both conditions are stated explicitly because they diverge for mixed bipartite states, which arise in the internal sub-factorizations of W (Section 6), where reduced states of subsystem pairs are generically mixed and can be mixed without being entangled.

Remark 2. Whether a formal Witness possesses phenomenal experience is a separate question entirely, addressed in Axiom V.

Witnessing is not an ontological event that physically destroys superpositions; it is the epistemological adoption of a local perspective on a globally pure state. The Witness cannot access the entire universe $\mathcal{H}_{\text{total}}$; it is restricted to its own local subsystem $\mathcal{H}_W$. To find the state of the Witness, we apply the partial trace, tracing out the rest of the universe:

$$\rho_W = \text{Tr}_R(|\Psi\rangle\langle\Psi|) = \sum_i p_i |w_i\rangle\langle w_i|. \quad (4)$$

Crucially, the partial trace does not describe what physically happens to the global wave function; it describes what the subsystem can know. Because of this epistemic restriction *together with the entanglement condition of Axiom III*, the von Neumann entropy of the Witness is strictly greater than zero:

$$S(\rho_W) = -\sum_i p_i \log p_i > 0. \quad (5)$$

(For a product state the restriction alone yields $S(\rho_W) = 0$; strict positivity is a consequence of the non-trivial Schmidt rank required by Axiom III.) Entropy is the intrinsic mathematical consequence of adopting a local perspective within an *entangled* system.

4 Axiom IV: Relational Time

Time is what conditional entropy accumulation looks like from inside a bipartition. If part of the Witness acts as a clock, we define the state of the Rest of the universe conditional on the clock reading a specific state $|w(\tau)\rangle$, following the Page–Wootters mechanism [4]¹:

Axiom 4 (Relational Time). *The emergent time parameter τ of a Witness is defined relationally: the (unnormalized) state of the environment conditional on the clock reading $|w(\tau)\rangle$ is*

$$|\tilde{\psi}(\tau)\rangle = (\langle w(\tau)|_C \otimes \mathbb{I}) |\Psi\rangle. \quad (6)$$

The parameter τ is the emergent, subjective time of the Witness; it labels the family of conditional states and carries no meaning external to the bipartition.

¹We use the standard shorthand $\langle w(\tau)|\Psi\rangle$ for the conditional state $(\langle w(\tau)|_C \otimes \mathbb{I}) |\Psi\rangle$, which is unnormalized as written; normalized conditional states are obtained by dividing by $\|(\langle w(\tau)|_C \otimes \mathbb{I}) |\Psi\rangle\|$. The Kuchař two-time-correlation objection to the Page–Wootters construction is resolved in the covariant formulation of Giovannetti, Lloyd, and Maccone [6]; for the full perspective-neutral formalism, see Höhn, Smith, and Lock [24].

4.1 The relational step and its entropy increment

A subtlety must be made explicit before entropy can be coupled to τ . Because $|\Psi\rangle$ is static (Axiom I), the reduced state $\rho_W = \text{Tr}_R(|\Psi\rangle\langle\Psi|)$ is a *fixed* operator: $S(\rho_W)$ is a single number and does not change. The entropy that accumulates along relational time is therefore not the entropy of the static reduced state, but the entropy of the Witness’s *records*, conditional on the clock reading. To formalize this, we endow the Witness with internal structure.

Definition 2 (Clock–memory decomposition, relational step, and entropy increment). Let the Witness factor as

$$\mathcal{H}_W = \mathcal{H}_C \otimes \mathcal{H}_M, \quad (7)$$

where C is a *clock* and M is a *memory*. The clock is non-interacting in the sense of Marletto–Vedral [5], securing the uniqueness of its factorization (Section 2); the memory is permitted to interact with R .

A *relational step* is a transition of the clock between consecutive mutually orthogonal clock states $|w(\tau_k)\rangle \perp |w(\tau_{k+1})\rangle$ —the natural unit of distinguishable relational change, and precisely the unit whose duration the Margolus–Levitin bound constrains.

The *entropy increment per relational step* is the change in the von Neumann entropy of the memory’s conditional state across one step:

$$\Delta S = S(\rho_M(\tau_{k+1})) - S(\rho_M(\tau_k)), \quad \rho_M(\tau) = \text{Tr}_R(|\psi_{MR}(\tau)\rangle\langle\psi_{MR}(\tau)|), \quad (8)$$

where $|\psi_{MR}(\tau)\rangle$ is the normalized conditional state of $M \otimes R$ given clock reading $|w(\tau)\rangle$. Throughout the remainder of the paper, ΔS denotes this conditional record-entropy increment, measured in bits.

Remark 3 (Finite clocks). The definition of a relational step idealizes a clock with exactly orthogonal reading states. Finite-dimensional clocks realize orthogonal readings only approximately; quantitative limits on the accuracy of autonomous finite-sized quantum clocks are given by Woods, Silva, and Oppenheim [9]. We work in the idealized limit and note that finite-clock corrections would soften, not eliminate, the bound derived below.

Remark 4 (Uniqueness, interaction, and the resolution of an apparent tension). The Marletto–Vedral uniqueness theorem [5] requires a non-interacting clock, while Landauer erasure is paradigmatically an interactive, dissipative process. These requirements would conflict if they were imposed on the same degree of freedom. The clock–memory decomposition (Definition 2) resolves the tension by separating the roles: the clock C is non-interacting, securing both the uniqueness of the temporal factorization and the validity of the Page–Wootters conditioning; the memory M interacts with R and bears the informational cost. Crucially, the entropy increment ΔS of Equation (8) is a *relational* property of the static entanglement structure of $|\Psi\rangle$, conditioned on clock readings—not dynamical dissipation unfolding in an external time. No external time parameter is invoked anywhere in the construction.

4.2 The optimal-witness time bound

To anchor τ physically, we compose two established results under a limiting assumption.

Landauer’s Principle [7] establishes the minimum energy cost of erasing (or irreversibly overwriting) ΔS bits of information at temperature T :

$$\Delta E \geq k_B T \ln 2 \cdot \Delta S. \quad (9)$$

The Margolus–Levitin quantum speed limit [8] bounds the minimum time for a quantum system with mean energy ΔE above its ground state to evolve to a distinguishable (orthogonal) state:

$$\Delta\tau \geq \frac{\pi\hbar}{2\Delta E}. \quad (10)$$

Note that direct substitution of the Landauer lower bound into the denominator of (10) does not yield a valid lower bound on $\Delta\tau$: a system consuming more energy than the Landauer minimum may tick faster. To obtain a definite relation between time and entropy, we model a thermodynamically optimal Witness:

Assumption 1 (Thermodynamic Optimality). The Witness processes information at the Landauer limit: the energy consumed per relational step is exactly the erasure cost of the records overwritten in that step,

$$\Delta E = k_B T \ln 2 \cdot \Delta S. \quad (11)$$

Implicit in this assumption is the identification of the record-entropy increment ΔS (Definition 2) with bits irreversibly overwritten in the finite memory M —i.e., the Witness is modeled as maintaining records of R in a bounded memory, so that each relational step requires overwriting prior records at Landauer cost.

Proposition 1 (Relational Time Step for an Optimal Witness). *Under Assumption 1, the duration of a relational step satisfies*

$$\Delta\tau_{\text{opt}} \geq \frac{\pi\hbar}{2k_B T \ln 2 \cdot \Delta S}. \quad (12)$$

Proof. Substitute the equality (11) into the Margolus–Levitin bound (10), applied to the orthogonalization of the clock state that constitutes one relational step (Definition 2). $\square$

Remark 5 (Physical interpretation). $\Delta\tau_{\text{opt}}$ characterizes the *slowest permissible clock* for a given entropy increment, not the fastest. A thermodynamically optimal Witness operates at the Landauer limit—the quasistatic regime of maximum efficiency and minimum energy expenditure. Any real (non-optimal) Witness dissipates $\Delta E > k_B T \ln 2 \cdot \Delta S$, and the Margolus–Levitin bound permits it to tick faster than $\Delta\tau_{\text{opt}}$. The bound is therefore a ceiling on the time step (an upper bound on how slowly a witness can tick per bit of entropy), not a floor. $\Delta\tau_{\text{opt}}$ is the maximum temporal cost per unit of relational entropy generation.

Remark 6 (The energy identification). This derivation identifies two physically distinct quantities: the dynamical energy $\langle \hat{H} \rangle - E_0$ driving unitary evolution in the Margolus–Levitin bound, and the dissipated heat $k_B T \ln 2 \cdot \Delta S$ of irreversible information erasure in the Landauer bound. The identification $\Delta E_{\text{ML}} = \Delta E_{\text{Landauer}}$ models a Witness whose entire available dynamical energy budget is committed to information processing at the Landauer limit—the quasistatic, thermodynamically optimal regime. This is a strong modeling assumption, not a derivation from first principles. The resulting bound is physically suggestive and dimensionally correct, but the strict equivalence between distinguishable-state evolution and information erasure remains heuristic in the general case.

Remark 7. Equation (12) entails that $\Delta\tau_{\text{opt}}$ is inversely proportional to both temperature and entropy generation. Hotter systems with greater entropy production have smaller $\Delta\tau_{\text{opt}}$ —meaning the ceiling on their time step is lower, permitting faster relational dynamics. Conversely, as $\Delta S \rightarrow 0$ (a nearly isolated Witness), $\Delta\tau_{\text{opt}} \rightarrow \infty$: the maximum time step diverges, and relational time freezes. This is consistent with the global stasis of the Wheeler–DeWitt equation (Axiom I)—a system with no entropy generation has no relational time.

Remark 8. This result is a quantum-information-theoretic analogue of the thermal time hypothesis of Connes and Rovelli [10], which proposes that the flow of time in generally covariant quantum theories emerges from the modular automorphism of the thermal state. Both frameworks anchor temporal flow to thermodynamic structure; the present derivation does so via information-erasure bounds rather than modular theory.

4.3 The temperature T

The temperature T appearing in (12) requires careful justification: because the global state is pure ($\hat{H}|\Psi\rangle = 0$), the reduced state ρ_W does not automatically carry a physical temperature. Two routes are available:

The modular route. Define T via the modular Hamiltonian $K_W = -\ln \rho_W$, following the algebraic framework of Connes and Rovelli [10]. This definition is intrinsic to ρ_W and requires no assumption about thermalization. However, the KMS temperature associated with the modular automorphism is $\beta = 1$ by construction in modular time; it acquires nontrivial physical content only when compared to an independent time flow. Since time is itself defined by Axiom IV using T , there is a circularity risk: the modular temperature requires a time flow for physical interpretation, but time requires T . This circularity is the same one that lives within the Connes–Rovelli thermal time hypothesis itself. Additionally, the modular Hamiltonian requires ρ_W to be faithful (no zero eigenvalues), which may not hold for all physically relevant states.

The ETH route. Invoke the Eigenstate Thermalization Hypothesis (ETH) [11], under which R acts as a thermalizing bath for W and the reduced state ρ_W is approximately thermal at a well-defined physical temperature. We emphasize that the version of ETH invoked here is its *spectral* formulation—a static statement about the matrix elements of local observables in the energy eigenbasis of $\hat{H}$ —so no external time parameter is presupposed. This route avoids the modular circularity by grounding T in the spectral properties of $\hat{H}$ restricted to the relevant energy shell, but introduces an additional physical assumption about the thermalization properties of the global Hamiltonian.

Both routes yield a consistent framework. We regard the choice between them as open and note that the framework’s structural conclusions—the coupling between entropy generation and relational time—hold under either interpretation of T .

5 Axiom V: Locating the Hard Problem

We possess a physical system generating epistemic entropy ($\Delta S > 0$) and a relational parameter tracking it (τ). However, this does not explain why τ is phenomenologically experienced as a flowing *now*.

The contribution of this framework is not to solve the hard problem of consciousness, but to *locate* it with mathematical precision—to define the structural constraints that any future solution must satisfy.

We first fix the physical category. Let **Phys** be the category whose objects are quantum states ρ_W of witness-candidate systems (with a fixed internal factorization, Section 6) and whose morphisms are channels implementable by local operations and classical communication (LOCC); composition is channel composition and identities are the identity channels. Because LOCC cannot increase entanglement, the witnessing states—those with $E_N(W:R) > 0$, per Axiom III—form a full subcategory **Phys**₊ that is *not* closed under LOCC: a morphism may carry an entangled state to a separable one. We regard this as structurally meaningful rather than defective: morphisms

that exit $\mathbf{Phys}_+$ model precisely the physical transitions that extinguish witnessing (compare the anesthesia transition of Section 7).

Axiom 5 (The Functor Constraint). *Let $\mathbf{Phen}$ be the (as yet formally undefined) category of phenomenal experience. The hard problem of consciousness is the joint problem of:*

- (i) *defining the category $\mathbf{Phen}$ —its objects (experiential states) and morphisms (transitions or relations between experiences); and*
- (ii) *constructing a structure-preserving functor F mapping the physical generation of entropy into the subjective generation of experience:*

$$F : \mathbf{Phys} \longrightarrow \mathbf{Phen}, \quad (13)$$

such that entropy-generating LOCC morphisms in $\mathbf{Phys}_+$ are carried to now-generating morphisms in $\mathbf{Phen}$, and morphisms exiting $\mathbf{Phys}_+$ are carried to morphisms extinguishing experience.

The framework does not presume that only F remains to be defined; the formalization of $\mathbf{Phen}$ is equally (and perhaps more fundamentally) open. What the framework provides is the constraint that F must be a functor—it must preserve the compositional structure of LOCC morphisms in $\mathbf{Phys}$ —and that it must factor through the entanglement topology of W (Section 6).

While Integrated Information Theory (IIT) [12] attempts to quantify the physical correlates of consciousness via the integrated information Φ of a system’s causal architecture, the Relational Witnessing Framework proposes a different structural approach: that F factors through a topological invariant of the entanglement structure internal to W .

Remark 9 (Necessary but not sufficient). Non-zero entanglement ($E_N(W:R) > 0$) is a formally necessary condition for witnessing within this framework, but it is not claimed to be sufficient for phenomenal experience. A single qubit entangled with a photon satisfies Axiom III but is not thereby conscious. The emergence of phenomenal experience likely requires sufficient topological complexity within the entanglement complex K_W —for instance, nontrivial persistent homology in dimension $k \geq 1$ (Section 6). The framework constrains the form that any sufficiency condition must take, but does not yet classify the threshold of topological complexity required.

6 The Topological Hypothesis: Formalizing G_W

We now define the entanglement graph G_W and its topological invariants. This section is conjectural and constitutes a research program, not a derivation.

Relation to prior work. The construction of simplicial complexes from multipartite quantum states, with pairwise entanglement or correlation measures as filtration parameters, is an established direction in topological data analysis. Di Pierro, Mancini, Memarzadeh, and Mengoni [18] first proposed the use of bipartite entanglement as a distance for Vietoris–Rips filtrations on multipartite qubit states; Mengoni, Memarzadeh, Di Pierro, and Mancini [19] developed the approach further. Olsthoorn [20] constructed filtered simplicial complexes from inverse quantum mutual information and applied persistent homology to condensed-matter spin systems. Hamilton and Leditzky [21] refined the construction using Čech complexes weighted by Tsallis-deformed total correlation, defining an Integrated Euler Characteristic as a topological summary of multipartite entanglement; operational interpretations of this summary via distillable-entanglement bounds have

subsequently been developed [22]. The construction given below specializes this general program to logarithmic negativity as the pairwise weight (motivated by Axiom III, where mixedness of the reduced state is the generic case), adopts the flag complex for its face-closure property and computational tractability, and embeds the resulting signature within the broader categorical framework of Axiom V—specifically, as the invariant through which the functor F is hypothesized to factor (Section 5). The application domain (relational time and phenomenal experience) is also distinct: prior work has studied abstract multipartite states and condensed-matter spin systems, not the use of entanglement topology as a constraint on theories of subjective experience.

6.1 Internal Factorization of W

For the entanglement structure internal to W to be well-defined, $\mathcal{H}_W$ must factor into subsystems:

$$\mathcal{H}_W = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n. \quad (14)$$

The same factorization-dependence issue identified by Zanardi et al. [2] applies here. We resolve it by recursive application of the algebraic criterion from Section 2: the preferred internal factorization is the one with respect to which the Witness’s internal observables act locally. In the emergent-time regime, this coincides with the sub-factorization stabilized by internal decoherence, as in Zurek’s einselection program [3].

6.2 The Entanglement Complex K_W

Definition 3 (Entanglement Complex). Given the internal factorization (14) and the reduced state ρ_W , we define the *entanglement complex* K_W as a flag complex (clique complex) constructed from the weighted 1-skeleton of pairwise entanglement:

- *0-simplices (vertices)*: individual subsystems $\{1, 2, \dots, n\}$.
- *1-simplices (edges)*: pairs (i, j) with weight given by the pairwise logarithmic negativity

$$w_{ij} = E_N(i:j) = \log \|\rho_{ij}^{T_i}\|_1, \quad (15)$$

where $\rho_{ij} = \text{Tr}_{\{i,j\}^c}(\rho_W)$ is the reduced state of the pair (obtained by tracing out all subsystems *other than* i and j) and T_i denotes partial transpose with respect to subsystem i . An edge (i, j) is present if and only if $w_{ij} > 0$.

- *k -simplices ($k \geq 2$)*: a subset $\{i_1, \dots, i_{k+1}\}$ is included as a k -simplex if and only if all $\binom{k+1}{2}$ pairwise edges (i_a, i_b) are present. The weight of a higher simplex is the minimum of its edge weights:

$$w(\sigma) = \min_{\{a,b\} \subset \sigma} w_{ab}. \quad (16)$$

Equivalently: for each threshold $\varepsilon > 0$, the complex K_W^ε is the flag complex of the graph G_W^ε whose edges are the pairs with $w_{ij} \geq \varepsilon$. The min-weight rule (16) makes the two descriptions coincide and guarantees that the thresholded complexes are nested: $\varepsilon' \leq \varepsilon \implies K_W^{\varepsilon'} \subseteq K_W^\varepsilon$.

The choice of logarithmic negativity is consistent with Axiom III: it detects genuine quantum entanglement and is zero for separable states. The flag complex construction is adopted for two reasons. First, it guarantees the face-closure property required of a valid simplicial complex: every face of an included simplex is itself included. An alternative construction weighting higher simplices by genuine multipartite entanglement measures (e.g., the n -tangle [13] or genuine multipartite

negativity [14]) can violate this property—a GHZ state of three qubits, for instance, possesses maximum tripartite entanglement but zero pairwise entanglement after partial trace, yielding a 2-simplex without its 1-simplex faces. Second, the flag complex is entirely determined by the 1-skeleton and is computationally tractable, since computing genuine multipartite entanglement measures for $k \geq 2$ is in general NP-hard for large systems.

Remark 10. The flag complex may fail to capture genuinely multipartite entanglement that exists independently of pairwise correlations; this limitation is intrinsic to any construction derived from pair marginals and is well-known in the broader persistent-homology-of-entanglement literature [18, 21]. We regard a richer simplicial construction that accommodates such states—while preserving face-closure—as an important open problem in the formalization of K_W .

6.3 Topological Invariants via Persistent Homology

Rather than choosing an arbitrary threshold to binarize the weighted complex, we apply persistent homology [15, 16], which tracks how topological features—connected components (H_0), loops (H_1), voids (H_2), and higher-dimensional cavities (H_k)—appear and disappear as the filtration parameter ε sweeps from the maximum weight down to zero.

At threshold ε , include all simplices $\sigma \in K_W$ with weight $w(\sigma) \geq \varepsilon$, where higher-simplex weights are given by (16). As ε decreases, more simplices enter the complex, and the homology changes. Each topological feature is born at some ε_b and dies at some $\varepsilon_d < \varepsilon_b$, yielding an interval $[\varepsilon_d, \varepsilon_b)$.

Definition 4 (Topological Signature). The *topological signature* of the Witness is the collection of persistence diagrams across all homological dimensions:

$$\chi(K_W) = \{\text{Dgm}_k(K_W)\}_{k=0}^{\dim K_W}, \quad (17)$$

where each Dgm_k is a multiset of intervals $[\varepsilon_d, \varepsilon_b)$ recording the birth and death of k -dimensional features.

Persistent homology has the desired properties for our framework:

- (i) *Stability*: small perturbations in the entanglement weights produce small changes in the persistence diagrams (with respect to the bottleneck or Wasserstein distance [17]). This ensures robustness under minor fluctuations.
- (ii) *Sensitivity to structure*: qualitative changes in the entanglement topology (e.g., severing a cluster of connections, as under anesthesia) produce measurable changes in the persistence diagrams.
- (iii) *Computability*: persistent homology is algorithmically tractable for finite complexes.

6.4 Class vs. Content: The Two-Level Structure

The persistent homology framework naturally formalizes the distinction between the *class* of experience and its *content*.

Let $\mathcal{B}$ be the space of persistence diagrams, equipped with the Wasserstein metric d_W . For each point $b \in \mathcal{B}$, define the *level set*:

$$F_b = \{\rho_W \mid \chi(K_W[\rho_W]) = b\}, \quad (18)$$

the set of all Witness states whose entanglement complex has topological signature b .

Hypothesis 1 (Two-Level Structure). *The functor F decomposes as:*

$$F(\rho_W) = F'(\chi(K_W), \rho_W), \quad (19)$$

where:

- $\chi(K_W) \in \mathcal{B}$ determines the structural class of experience (e.g., wakefulness vs. dreamless sleep vs. anesthesia);
- the specific state ρ_W within the level set $F_{\chi(K_W)}$ determines the phenomenal content (e.g., seeing red vs. hearing a tone).

Diagrammatically, the hypothesis is that F factors as

$$\begin{array}{ccc} \mathbf{Phys} & \xrightarrow{F} & \mathbf{Phen} \\ & \searrow (\chi, \text{id}) \quad \nearrow F' & \\ & \mathcal{B} \times \mathbf{Phys} & \end{array}$$

We label this a hypothesis rather than a proposition: since F has not been constructed (Axiom V), no property of F can currently be proven; the decomposition (19) is a structural constraint the framework imposes on any candidate F .

This structure is that of a surjection $\pi : E \rightarrow \mathcal{B}$ with level sets, where E is the total space of Witness states, $\mathcal{B}$ is the base space of topological signatures, and each $F_b = \pi^{-1}(b)$ is the set of states sharing that signature. The map $\rho_W \mapsto \chi(K_W)$ is continuous: logarithmic negativity is continuous in the trace norm, so the entanglement weights w_{ij} vary continuously with ρ_W , and the stability theorem for persistent homology [17] then makes the persistence diagrams Lipschitz with respect to sup-norm perturbations of the filtration values. Whether the level sets F_b are locally homeomorphic to each other—the condition required for a genuine fiber bundle—is an open question. If local triviality holds, π acquires the full structure of a fiber bundle; we conjecture but do not prove this.

The Topological Hypothesis, as developed through Sections 6.1–6.4, provides the following: a well-defined simplicial complex K_W encoding the entanglement structure of the Witness, a stable topological invariant $\chi(K_W)$ via persistent homology, and a two-level decomposition of the functor F into structural class and phenomenal content. Whether the space of topological signatures $\mathcal{B}$ itself has interesting higher homotopy structure—and in particular whether such structure might exhibit periodicity—is an open problem explored in Appendix A.

7 Structural Consequences and Empirical Constraints

We outline three structural consequences of the framework. The first two are interpretive—they follow from the formalism but do not predict deviations from standard quantum mechanics. The third is in principle empirically distinguishable, subject to the caveats stated.

1. **Relational time freezing.** For a perfectly isolated quantum system ($\Delta S \rightarrow 0$), Equation (12) yields $\Delta\tau_{\text{opt}} \rightarrow \infty$: relational time ceases. This is not a temporal anomaly measurable against an external clock; it is the framework’s statement that a system generating no conditional record entropy has no relational time parameter at all. This is consistent with the global stasis of the Wheeler–DeWitt equation (Axiom I) and with the Page–Wootters mechanism, in which time is strictly relational. We note that this consequence is interpretive:

it does not predict any observable deviation from the Schrödinger equation’s predictions for isolated systems. The framework is empirically equivalent to standard quantum mechanics in this regime; the difference is ontological, not operational.

2. **The entanglement prerequisite for experience.** A system completely isolated into a pure state ($S = 0$, $E_N = 0$) cannot, by definition, possess a temporal *now* or act as a formal Witness. Genuine quantum entanglement is a strict physical prerequisite for phenomenal experience within this framework.
3. **Topological transitions under anesthesia.** If the class of experience is determined by the persistence diagrams $\chi(K_W)$, then the transition from wakefulness to anesthesia-induced unconsciousness should correspond to a measurable topological transition in the entanglement structure—specifically, a collapse in the persistent homology (loss of long-lived topological features in H_1 and higher). Two caveats are essential. First, the framework’s invariant is defined on quantum entanglement structure, while existing empirical consciousness indices—most notably the perturbational complexity index of Casali et al. [23]—are classical complexity measures; they can serve at best as proxies for the quantity the framework specifies. Second, the test presupposes that the witness’s physical substrate instantiates measurable entanglement structure at the relevant scales; whether this holds for biological systems is an open empirical question that the framework does not settle. Within those caveats, the prediction is in principle testable via quantum-information-theoretic analysis of the relevant substrate, extending existing work connecting complexity measures to consciousness [23].

8 The Synthesized Formalism

The Relational Witnessing Framework proposes that witnessing, entropy, and time are three descriptions of a single relational structure. We now make this claim precise. The three concepts are co-generated by a single structural choice—the epistemic bipartition $\mathcal{H}_W \otimes \mathcal{H}_R$ —through its two canonical maps. The bipartition induces:

- via the partial trace Tr_R , the epistemic boundary of the Witness (what the subsystem can access) and the mixed state ρ_W whose von Neumann entropy $S(\rho_W) > 0$ is the entanglement entropy; and
- via conditionalization on clock states $|w(\tau)\rangle$, the relational time parameter τ and the record-entropy increments ΔS that pace it (Definition 2).

The identity claim is that these are not three independently motivated concepts that happen to coincide, but three projections of the structure induced by a single cut: both the partial trace and the conditioning map are canonically determined by the bipartition, with no further choices beyond the clock–memory decomposition made explicit in Definition 2. A stronger formulation would require proving that witnessing, entropy, and time are isomorphic as objects in some encompassing category—that there exists a formal sense in which any one determines the other two. We regard this stronger claim as a conjecture motivated by the co-generation structure, not a proven theorem.

The synthesized formalism is:

$$\hat{H} |\Psi\rangle = 0 \quad (\text{The Timeless Whole}) \quad (20)$$

$$\mathcal{H}_{\text{total}} = \mathcal{H}_W \otimes \mathcal{H}_R \quad (\text{The Algebraic Bipartition}) \quad (21)$$

$$\rho_W = \text{Tr}_R(|\Psi\rangle\langle\Psi|), \quad E_N(W:R) > 0 \quad (\text{The Epistemic Cut}) \quad (22)$$

$$\Delta\tau_{\text{opt}} \geq \frac{\pi\hbar}{2k_B T \ln 2 \cdot \Delta S} \quad (\text{Relational Time}) \quad (23)$$

$$F(\rho_W) = F'(\chi(K_W), \rho_W) \quad (\text{The Topological Constraint}) \quad (24)$$

9 Conclusion

This framework makes a specific structural claim: that time, entropy, and witnessing are not three separate phenomena requiring three separate explanations, but three projections of a single relational structure induced by the epistemic bipartition of a timeless whole—through the partial trace and clock conditioning it canonically determines. The claim of co-generation is demonstrated; the stronger claim of formal isomorphism remains an open conjecture. The deductive core (Axioms I–IV) rests on established physics: the Wheeler–DeWitt equation, the Page–Wootters mechanism, Landauer’s principle, and the Margolus–Levitin speed limit, composed under two explicit modeling assumptions—the thermodynamic optimality of the Witness (Assumption 1) and the clock–memory decomposition with its identification of record-entropy increments as Landauer-erased bits (Definition 2). The conjectural extension (Axiom V and the Topological Hypothesis) proposes that the hard problem of consciousness can be *located*—if not yet solved—as the joint problem of defining both the category of phenomenal experience and a structure-preserving functor from the category of witness states into it, and that this functor factors through the persistent homology of the entanglement complex.

We acknowledge the most dangerous foundational objection directly: the framework’s conclusions depend on the choice of tensor product structure (the bipartition), and different factorizations yield different witnesses, entropies, and time parameters. If the preferred factorization is not uniquely determined, the framework’s outputs may be artifacts of a decomposition choice rather than features of reality. We have offered an algebraic criterion (observable-induced factorization), strengthened to uniqueness for the non-interacting clock factor by the Marletto–Vedral theorem [5], and supplemented by a dynamical consistency check (Quantum Darwinism); but whether this fully resolves the ambiguity for the remaining factors—in particular the memory factor and the internal factorization of Section 6.1—is an open question. The framework stands or falls on the defensibility of its factorization criterion.

The framework connects naturally to several active research programs: the quantum reference frame formalism of Höhn, Smith, and Lock [24], which provides a rigorous foundation for the Page–Wootters mechanism; the thermal time hypothesis of Connes and Rovelli [10], which similarly anchors temporal flow to thermodynamic structure; the $ER = EPR$ program of Maldacena and Susskind [25] and Van Raamsdonk’s entanglement–geometry correspondence [26], which suggest that entanglement entropy generates spatial geometry as well as time; and topological entanglement entropy in the sense of Kitaev and Preskill [27], which provides precedent for extracting topological invariants from entanglement structure.

Whether the functor F can be explicitly constructed, and whether the space of topological signatures exhibits further structure (see Appendix A), are open mathematical questions. What the Relational Witnessing Framework provides is the precise scaffold—the categorical, topological, and information-theoretic constraints—that any such construction must satisfy.

A Open Problem: Periodicity in the Entanglement Topology

This appendix poses a speculative but precise open problem motivated by the topological framework of Section 6. It is structurally independent of the main results: the framework’s deductive core (Axioms I–IV) and the Topological Hypothesis (Section 6, through the two-level structure of Section 6.4) do not depend on the conjecture stated here.

The mathematician Raoul Bott identified a deep periodicity in the homotopy groups of the classical groups: $\pi_{k+8}(O) \cong \pi_k(O)$ and $\pi_{k+2}(U) \cong \pi_k(U)$ [28]. This periodicity governs the classification of fiber bundles and the structure of K-theory. We ask whether an analogous periodicity appears in the topology of entanglement.

Conjecture 1 (Bott Periodicity in Entanglement Topology). *Let $\mathcal{B}_n$ denote the space of topological signatures $\chi(K_W)$ for Witnesses with n internal subsystems. As n increases, the homotopy groups $\pi_k(\mathcal{B}_n)$ exhibit periodicity:*

$$\pi_{k+p}(\mathcal{B}_n) \cong \pi_k(\mathcal{B}_n) \quad \text{for some fixed period } p, \quad (25)$$

reflecting a fundamental periodicity in the structural classes of entanglement topology available to witnessing systems of increasing complexity.

If true, this would imply that the menu of structurally distinct experience-classes is not unbounded but recurs with a characteristic period—that beyond a certain complexity, new subsystems do not generate fundamentally new topological classes, but repeat existing ones at higher resolution.

The natural route to establishing Conjecture 1 would be to show that the entanglement complexes K_W , equipped with the local Hilbert spaces $\mathcal{H}_i$ as fibers over their vertices, define complex vector bundles. Rank- n complex vector bundles are classified by maps to $BU(n)$; passing to the stable limit $BU = \text{colim}_n BU(n)$, Bott periodicity takes the form $\pi_{k+2}(BU) \cong \pi_k(BU)$ for $k \geq 1$. Analogous applications of K-theoretic periodicity to physical quantum systems have been developed, most notably in Kitaev’s periodic-table classification of topological insulators and superconductors [29], which grounds the plausibility—though not the validity—of seeking such periodicity in the space of entanglement signatures. If a map from entanglement complexes to BU can be constructed, the periodicity of $\pi_k(\mathcal{B}_n)$ would follow from the periodicity of $\pi_k(BU)$ via the induced homomorphism on homotopy groups.

Two significant obstacles must be acknowledged:

1. **Transition functions.** A collection of vector spaces over vertices does not constitute a vector bundle without transition functions along the edges that glue the fibers together. To rigorously map K_W to $BU(n)$, future work must define these transition functions—potentially utilizing the local modular flow or the entanglement Hamiltonian $K_{ij} = -\ln \rho_{ij}$ to provide the requisite holonomy between $\mathcal{H}_i$ and $\mathcal{H}_j$ along each 1-simplex. Without such transition functions, the bundle is trivial and the conjecture is void.
2. **Contractibility.** The space of persistence diagrams equipped with the Wasserstein or bottleneck metric is known to be contractible for certain natural filtrations [30]. If $\mathcal{B}_n$ is contractible, then $\pi_k(\mathcal{B}_n) = 0$ for all $k \geq 1$, and the periodicity conjecture is trivially satisfied ($0 \cong 0$) but content-free. The conjecture has nontrivial content only if the relevant subspace of $\mathcal{B}_n$ —the persistence diagrams actually realizable by entanglement complexes of n -partite quantum states—is non-contractible. Determining the homotopy type of this physically realizable subspace is an open problem at the intersection of topological data analysis, K-theory, and quantum information theory.

We state this conjecture because it is precise, falsifiable, and—if true—would have profound consequences for the structure of the functor F . But we separate it from the main body of the paper to make clear that the Relational Witnessing Framework does not depend on it.

References

- [1] B. S. DeWitt, “Quantum Theory of Gravity. I. The Canonical Theory,” *Phys. Rev.* **160**, 1113 (1967).
- [2] P. Zanardi, D. A. Lidar, and S. Lloyd, “Quantum tensor product structures are observable induced,” *Phys. Rev. Lett.* **92**, 060402 (2004).
- [3] W. H. Zurek, “Quantum Darwinism,” *Nature Phys.* **5**, 181 (2009).
- [4] D. N. Page and W. K. Wootters, “Evolution without evolution: Dynamics described by stationary observables,” *Phys. Rev. D* **27**, 2885 (1983).
- [5] C. Marletto and V. Vedral, “Evolution without evolution and without ambiguities,” *Phys. Rev. D* **95**, 043510 (2017), arXiv:1610.04773.
- [6] V. Giovannetti, S. Lloyd, and L. Maccone, “Quantum time,” *Phys. Rev. D* **92**, 045033 (2015).
- [7] R. Landauer, “Irreversibility and Heat Generation in the Computing Process,” *IBM J. Res. Dev.* **5**, 183 (1961).
- [8] N. Margolus and L. B. Levitin, “The maximum speed of dynamical evolution,” *Physica D* **120**, 188 (1998).
- [9] M. P. Woods, R. Silva, and J. Oppenheim, “Autonomous quantum machines and finite-sized clocks,” *Ann. Henri Poincaré* **20**, 125 (2019).
- [10] A. Connes and C. Rovelli, “Von Neumann algebra automorphisms and time–thermodynamics relation in generally covariant quantum theories,” *Class. Quantum Grav.* **11**, 2899 (1994).
- [11] M. Srednicki, “Chaos and quantum thermalization,” *Phys. Rev. E* **50**, 888 (1994).
- [12] G. Tononi, M. Boly, M. Massimini, and C. Koch, “Integrated information theory: from consciousness to its physical substrate,” *Nature Rev. Neurosci.* **17**, 450 (2016).
- [13] V. Coffman, J. Kundu, and W. K. Wootters, “Distributed entanglement,” *Phys. Rev. A* **61**, 052306 (2000).
- [14] B. Jungnitsch, T. Moroder, and O. Gühne, “Taming multiparticle entanglement,” *Phys. Rev. Lett.* **106**, 190502 (2011).
- [15] H. Edelsbrunner, D. Letscher, and A. Zomorodian, “Topological persistence and simplification,” *Discrete Comput. Geom.* **28**, 511 (2002).
- [16] A. Zomorodian and G. Carlsson, “Computing persistent homology,” *Discrete Comput. Geom.* **33**, 249 (2005).
- [17] D. Cohen-Steiner, H. Edelsbrunner, and J. Harer, “Stability of persistence diagrams,” *Discrete Comput. Geom.* **37**, 103 (2007).

- [18] A. Di Pierro, S. Mancini, L. Memarzadeh, and R. Mengoni, “Homological analysis of multi-qubit entanglement,” *Europhys. Lett.* **123**, 30006 (2018), arXiv:1802.04572.
- [19] R. Mengoni, L. Memarzadeh, A. Di Pierro, and S. Mancini, “Persistent homology analysis of multiqubit entanglement,” arXiv:1907.06914 (2019).
- [20] B. Olsthoorn, “Persistent homology of quantum entanglement,” *Phys. Rev. B* **107**, 115174 (2023), arXiv:2110.10214.
- [21] G. A. Hamilton and F. Leditzky, “Probing multipartite entanglement through persistent homology,” *Commun. Math. Phys.* **405**, 125 (2024), arXiv:2307.07492.
- [22] C. Spellman et al., “Interpreting Multipartite Entanglement through Topological Summaries,” arXiv:2505.00642 (2025).
- [23] A. G. Casali et al., “A theoretically based index of consciousness independent of sensory processing and behavior,” *Sci. Transl. Med.* **5**, 198ra105 (2013).
- [24] P. A. Höhn, A. R. H. Smith, and M. P. E. Lock, “Trinity of relational quantum dynamics,” *Phys. Rev. D* **104**, 066001 (2021).
- [25] J. Maldacena and L. Susskind, “Cool horizons for entangled black holes,” *Fortschr. Phys.* **61**, 781 (2013).
- [26] M. Van Raamsdonk, “Building up spacetime with quantum entanglement,” *Gen. Rel. Grav.* **42**, 2323 (2010).
- [27] A. Kitaev and J. Preskill, “Topological entanglement entropy,” *Phys. Rev. Lett.* **96**, 110404 (2006).
- [28] R. Bott, “The stable homotopy of the classical groups,” *Ann. of Math.* **70**, 313 (1959).
- [29] A. Kitaev, “Periodic table for topological insulators and superconductors,” *AIP Conf. Proc.* **1134**, 22 (2009), arXiv:0901.2686.
- [30] Y. Mileyko, S. Mukherjee, and J. Harer, “Probability measures on the space of persistence diagrams,” *Inverse Problems* **27**, 124007 (2011).